

Phishing Email Detection & Awareness Analysis



Cyber Security Task 2 – Internship Report (2026)

Program: Future Interns — Cyber Security Internship

Task: Task 2 — Phishing Email Detection & Awareness Analysis

1. Introduction

Phishing is a common social-engineering technique in which attackers imitate trusted organizations or situations to persuade users to reveal information, follow a link, or take an unsafe action. This task focuses on recognizing those warning signs through four controlled email samples.

The assessment reviews the sender, wording, domain, URL, requested action, and level of pressure used in each message. The aim is to identify the likely threat and explain the safe response in simple terms.

2. Objectives

- Identify common phishing and social-engineering indicators.
- Review sender addresses, domains, subjects, and links.
- Classify each sample according to its potential risk.
- Explain the likely impact in clear language.
- Provide practical phishing-prevention guidance.

3. Investigation Approach

Stage	What was reviewed
Email review	Sender, subject, message wording and requested action
Domain review	Whether the sender/domain matches the claimed organization
URL review	Destination domain and purpose of the supplied link
Indicator analysis	Urgency, impersonation, sensitive requests and other warning signs
Risk assessment	Likely objective and potential impact

Note: The supplied .eml samples contain basic From, To and Subject information but do not provide complete delivery headers such as Received, Return-Path, SPF, DKIM or DMARC results. Therefore, this report does not claim full mail-server authentication analysis.

4. Tools & Materials

- Email (.eml) files for controlled analysis
- Browser-based inspection of domains and URLs

- Microsoft Word / PDF for documentation
- Git and GitHub for project submission

Phishing Email Analysis

Sample 1 — Microsoft MFA Verification

Subject: Urgent: MFA Verification Required Today

Sender: security@microsoft-account-alert.invalid

The email claims that an unusual Microsoft 365 sign-in was detected and asks the recipient to complete MFA verification within 24 hours.

Indicator	Observation
Brand impersonation	Uses Microsoft 365 in the sender name and message context.
Suspicious domain	The sender domain is not Microsoft's normal domain.
Urgency	A 24-hour deadline pressures the recipient to act quickly.
Verification request	The recipient is directed to an external verification page.
Suspicious URL	The link uses an unrelated .invalid domain.

Risk Verdict: HIGH — PHISHING

Likely objective: Steal account credentials or other Microsoft 365 account information through a fake verification process.

Sample 2 — Bank Transaction Verification

Subject: Immediate Action Required: Transaction Verification

Sender: fraud-alert@securebank-notice.invalid

The message reports an INR 32,750 transaction and asks the recipient to confirm account details immediately.

Indicator	Observation
Financial lure	A specific transaction amount is used to

	create concern.
Urgency	The recipient is told to verify the transaction immediately.
Generic greeting	The message addresses the recipient only as 'Dear Customer'.
Suspicious domain	The sender domain does not represent a verified bank domain.
Verification link	The supplied URL points to an unrelated external domain.
Sensitive context	The email refers to account details, OTP and PIN.

Risk Verdict: HIGH — PHISHING

Likely objective: Obtain banking credentials or other financial information.

Sample 3 — Payroll Profile Update

Subject: Payroll Profile Update Required

Sender: payroll@company-benefits-update.invalid

The email claims that a payroll system update requires the employee to confirm payment information before a processing deadline.

Indicator	Observation
Payroll lure	Payment information is used to make the request appear important.
Generic greeting	The recipient is addressed as 'Dear Employee'.
Deadline pressure	The message refers to a payroll processing deadline.
Suspicious domain	The sender domain is not tied to a verified employer.
External link	The recipient is directed to an unfamiliar update page.

Risk Verdict: HIGH — PHISHING

Likely objective: Collect employee, payroll, or financial information.

Sample 4 — Shared Compensation Document

Subject: A Document Has Been Shared With You

Sender: shared-file@cloud-document-alert.invalid

The email claims that an 'Annual Compensation Review.pdf' has been shared and asks the recipient to open a secure viewer.

Indicator	Observation
Unexpected document	The recipient was not given context for the shared file.
Sensitive topic	Compensation information is used to attract attention.
Suspicious sender	The domain does not correspond to a verified document service.
External link	The message directs the recipient to an unfamiliar viewer.
Limited availability	The document is described as available for a limited time.

Risk Verdict: HIGH — PHISHING

Likely objective: Redirect the recipient to a fraudulent page and potentially obtain account information.

5. Risk Summary

Sample	Primary Threat	Risk
Microsoft MFA Verification	Credential Phishing	● High
Bank Transaction Verification	Financial Phishing	● High
Payroll Profile Update	Information Theft	● High
Shared Compensation Document	Credential Phishing	● High

Security Awareness & Prevention

Key findings and recommendations

6. Common Phishing Indicators

- Urgent deadlines designed to create panic or reduce careful verification.
- Sender domains that do not match the organization being claimed.
- Generic greetings instead of clear recipient identification.
- Requests to verify accounts, payment information, or other sensitive data.
- Links leading to unfamiliar or unrelated domains.
- Unexpected documents or messages designed to attract curiosity.

7. Prevention Guidelines

Do	Don't
Verify the complete sender address.	Do not trust the display name alone.
Open important services through their official website/app.	Do not use unexpected login or verification links.
Verify unusual requests through a trusted channel.	Do not share passwords, OTPs or PINs.
Enable Multi-Factor Authentication.	Do not act simply because an email says 'urgent'.
Report suspicious messages to IT/security.	Do not provide personal or financial details without verification.

8. Practical Response

If a suspicious email is received, stop before clicking anything. Check the sender and domain, inspect the destination of the link without opening it, and verify the request through an independently known official channel. If the message remains suspicious, report it to the appropriate security or IT team.

9. Conclusion

The four samples demonstrate how phishing attempts combine familiar brands, financial concerns, workplace processes, urgency, and external links to influence user behaviour. Reviewing these indicators systematically makes it easier to recognize suspicious messages before information is disclosed.

The task provided practical experience in phishing detection, social-engineering analysis, risk classification, and security awareness documentation.

Author

Rahul Reghu Rajan | BCA | Cybersecurity | CEH

Safety Note

The samples used for this task are controlled educational examples. Their URLs use non-operational .invalid domains and are intended only for cybersecurity training and awareness.